

SS1 Physics Diagnostic (Part 2)

Curriculum: Cambridge IGCSE / O-Level

Cluster: Waves, Light, Electricity & Magnetism (Questions 21–40)

No.	Question	Ans	Diagnostic Focus
21	The frequency of a wave is 50 Hz. Its period is: A) 0.02 s B) 2 s C) 50 s D) 100 s	A	Wave relationships
22	Which wave can travel through a vacuum? A) Sound B) Water C) Light D) Seismic	C	Electromagnetic waves
23	Reflection occurs when light: A) Changes speed B) Bounces off a surface C) Splits into colours D) Passes through glass	B	Light
24	The image formed by a plane mirror is: A) Real B) Inverted C) Virtual D) Magnified	C	Image formation
25	Unit of electric current: A) Volt B) Ampere C) Ohm D) Watt	B	Electrical quantities
26	Using $V=IR$, find I when $V=12\text{ V}$ and $R=4\ \Omega$. A) 2 A B) 3 A C) 4 A D) 48 A	B	Ohm's law
27	Resistors in series have: A) Lower total resistance B) Equal voltage only C) Resistances added D) Constant current lost	C	Series circuits
28	A fuse protects a circuit by: A) Increasing current B) Melting when current is excessive C) Storing charge D) Reducing voltage	B	Electrical safety
29	Which material is a good conductor? A) Rubber B) Plastic C) Copper D) Wood	C	Conductors
30	The magnetic field around a straight current-carrying wire is: A) Circular B) Straight C) Elliptical D) Random	A	Magnetism
31	Transformer efficiency is highest when energy losses are: A) Maximum B) Zero/minimal C) Variable D) Unknown	B	Transformers
32	Sound cannot travel through: A) Solids B) Liquids C) Gases D) Vacuum	D	Sound waves
33	Higher frequency means: A) Lower pitch B) Higher pitch C) Greater loudness only D) Longer period	B	Sound properties
34	Refraction is caused by: A) Change in speed B) Reflection C) Diffraction D) Absorption	A	Light
35	Electrical power is given by: A) $P=IV$ B) $P=IR$ C) $P=VR$ D) $P=I/R$	A	Power
36	A 60 W bulb operates for 2 h. Energy used is: A)120 Wh B)30 Wh C)62 Wh D)600 Wh	A	Energy calculations

37	The SI unit of charge is: A) Coulomb B) Ampere C) Volt D) Farad	A	Charge
38	Which device converts electrical energy to mechanical energy? A) Generator B) Motor C) Transformer D) Cell	B	Applications
39	Electromagnetic induction occurs when: A) A magnet and conductor move relative to each other B) A battery is disconnected C) A resistor heats up D) Current stops permanently	A	Induction
40	Diagnostic insight: Weak performance on Questions 26, 35, 36 and 39 indicates difficulty with quantitative electricity and applied physics concepts expected in Cambridge SS1.	—	Readiness indicator